

Data Science

BSc Computer Science - Durham University

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1 Module Introduction

Data Science has the module code COMP2271 and is led by Professor Thomas Erlebach.

1.1 Assessment

This module is assessed through two pieces of summative coursework and two final exams, these can be seen in Table 1. **Final dates to be confirmed.**

Title	Opening	Due	Feedback	Professor	Weight
Data Cleaning & Analytics	Week 11	n/a	13/02/2027	Dr Igor Razgon	25%
Image Processing	16/01/2027	26/02/2027	26/03/2027	Dr George Koulrieris	25%
Probability and Graphics	May Exam	n/a	n/a	Razgon & Koulrieris	50%

Table 1: Assessments for Data Science.

1.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 2.

Day	Time	Class Type	Location	Lecturer
Monday	12:00 - 13:00	Lecture	W103	Dr Igor Razgon
Friday	10:00 - 11:00	Lecture	W103	Dr Igor Razgon
Friday	16:00 - 18:00	Computer Class	CC-0007	n/a

Table 2: Michaelmas Term Timetable for Data Science.

2 Probability and Statistics

This section is taught by Dr Igor Razgon. The recommended reading for this topic is:

- Probability and Statistics (Degroot, M.H. and Schervish, M.J., 2012).

2.1 Basics of Counting and Probability

2.2 Dependence, Conditional Probability, and Bayes' Theorem

2.3 Discrete Random Values and Probability Distributions

2.4 Continuous Probability Distributions

2.5 The Normal Distribution

2.6 Multivariate Distributions

2.7 Statistics and the Central Limit Theorem

2.8 Point Estimators, Confidence Intervals, and Hypothesis Testing

2.9 Hypothesis Testing

2.10 From Statistics to Data

3 Data Cleaning and Analytics

This section is taught by Dr Igor Razgon. **Check for recommended reading.**

4 Computer Graphics

This section is taught by Dr George Alex Koulieris. [Check for recommended reading.](#)

5 Image Processing

This section is taught by Dr George Alex Koulieris. [Check for recommended reading.](#)

Part I

Appendix

6 Bibliography

6.1 Module Professors

Module Lead: Professor Thoas Erlebach

- Focus: Algorithms and networks.
- Email: thomas.erlebach@durham.ac.uk
- Office Hours: n/a.

Lecturer: Doctor Igor Razgon

- Focus: Algorithms and complexity, graph theory, and discrete mathematics.
- Email: igor.razgon@durham.ac.uk
- Office Hours: Friday 14:00 - 15:00.

Lecturer: Doctor George Alex Koulieris

- Focus: Image processing and computer graphics.
- Email: georgios.a.koulieris@durham.ac.uk
- Office Hours: Please email to make an appointment.

6.2 References

6.3 List of Figures

6.4 Reading List

6.4.1 Machine Learning

Degroot, M.H. and Schervish, M.J. (2012). Probability and Statistics. Boston Mass.: Pearson.